Parth Nitesh Chaudhari

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Education

International Institute of Information Technology, Hyderabad — Hyderabad, India
B.Tech in Computer Science and Engineering — *July 2024 – May 2028*

Experience

Undergraduate Research Assistant — April 2026 – Present
Data Science and Analytics Centre, IIIT-H (Prof. Kamalakar Karlapalem) — *Python, JavaScript, OpenCV*

- Built a Python pipeline that turns player-tracking data from **17 professional matches** (A-League, Bundesliga) into **8,087** labelled phases of play, with 9 defensive-shape metrics sampled every 0.5 s.
- Built a framework-free Canvas/JavaScript dashboard that steps through **9,566** on-ball moments and labels **24,374** passing options as clear, contested or blocked; the blocking rule held on 5,803 held-out passes (**86% vs 62%** completion).

Software Engineering Team Project — January 2026 – May 2026
Cosmic 3D, Maker's Lab (client project, DASS course) — *TypeScript, React, Express, MQTT, ESP32*

- 5-person team building remote control and live monitoring for ESP32 3D printers (React, Express, Three.js, JWT). Owned the MQTT layer: configured Mosquitto over TCP and WebSockets and rewrote the backend MQTT service and dashboard subscriptions so printer telemetry reaches any browser on the network.

Projects

Event Management Platform | *React, Express, MongoDB, Socket.IO* — February 2026

- Built solo and deployed (Vercel + Render) a **7,800-line** full-stack fest platform with **52 REST endpoints**, 9 Mongoose models and JWT role-based portals for participants, organisers and admins.
- Socket.IO forums and team chat with file uploads, a custom registration-form builder, team matchmaking, payment-proof approval, QR e-tickets by email, double-blind feedback and reCAPTCHA-protected password reset.

CLI-Docs: Distributed Document Store | *C, pthreads, TCP sockets* — October 2025 – November 2025

- Wrote most of a **4,700-line** C system (2-person team, 18 of 22 commits): a name server, multiple multi-threaded storage servers and a CLI client, with clients reading and writing directly to storage servers.
- Per-sentence write locks let many users edit one document while reads continue; djb2 hash tables give O(1) user and file lookups; least-loaded placement, persisted metadata, one-step undo and word-by-word streaming.

xv6 Kernel: Schedulers and Demand Paging | *C, RISC-V, QEMU* — 2025

- Added FCFS and a Completely Fair Scheduler (nice −20 to +20) to xv6, plus demand paging with per-process FIFO eviction and swap slots, and benchmarked schedulers on CPU-bound and mixed I/O workloads.

C-Shell | *C, POSIX* — September 2025

- **1,100-line** POSIX shell with a grammar-based parser for pipelines, `;`, `&&`, `||`, `&` and `< > >>` redirection; job control (`fg`/`bg`, SIGINT/SIGTSTP), async child reaping and persistent history.

Network Protocol Suite | *C, UDP, libpcap, BPF* — September 2025 – October 2025

- TCP-like transport over UDP: 12-byte header, 3-way handshake and 4-way teardown, 10-packet sliding window, cumulative ACKs and flow control, with file transfer (MD5-verified) and `select()`-based chat.
- libpcap packet sniffer decoding Ethernet, IPv4/IPv6, ARP, TCP/UDP and HTTP/DNS ports, with BPF filters and a 10,000-packet in-memory store for offline hex/ASCII inspection.

Institute Management Android App | *Kotlin, Jetpack Compose, Material 3* — April 2026

- 2-person team; wrote the **5,500-line** Kotlin codebase (26 files): role-aware dashboards for students and faculty, OTP sign-in flow, 24-hour timetable with alerts and per-course attendance marking.

LLM Confidence Under Pressure | *PyTorch, Hugging Face, Slurm, W&B* — September 2026 – October 2026

- Reproducible evaluation harness running Qwen2.5-7B and Llama-3.1-8B (4-bit) through 6-turn conversations on 450 claims on a Slurm cluster; every report table rebuilds from logged runs without a GPU.

Technical Skills

Languages: C, C++, Python, TypeScript, JavaScript, Kotlin, SQL
Systems: POSIX processes and signals, pthreads, TCP/UDP sockets, libpcap, xv6, Linux
Web and Mobile: React, Node.js, Express, MongoDB, Socket.IO, MQTT, Jetpack Compose
Tools: Git, Make, QEMU, Slurm, Weights & Biases